Assignment 4

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Question 1-4

Numerical Solution of Differential Equation

Solve the differential equation

$$\frac{dy}{dx} = y^2 + 1$$

in the interval $x \in [0, 1.55]$ with initial condition $y(0) = 0$.

- Use Euler, Modified Euler, and Improved Euler methods with step size $dx = 0.001$.
- Use the RK4 method with step size $dx = 0.01$.
- Save the computed data in four separate files.

The exact solution is:

$$y(x) = \tan(x)$$

and the true value at $x = 1.55$ is:

$$y_A = 48.07848247921907$$

Let the numerical solutions at $x = 1.55$ be denoted as:

$$y_E, \quad y_{ME}, \quad y_{IE}, \quad y_{RK4}$$

Compute the error defined as:

$$\text{Error} = |y_A - y_{\text{method}}|$$

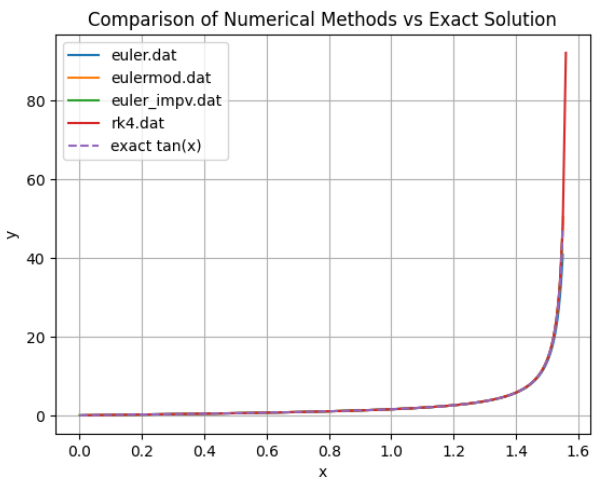


Figure 1: Comparison of numerical methods

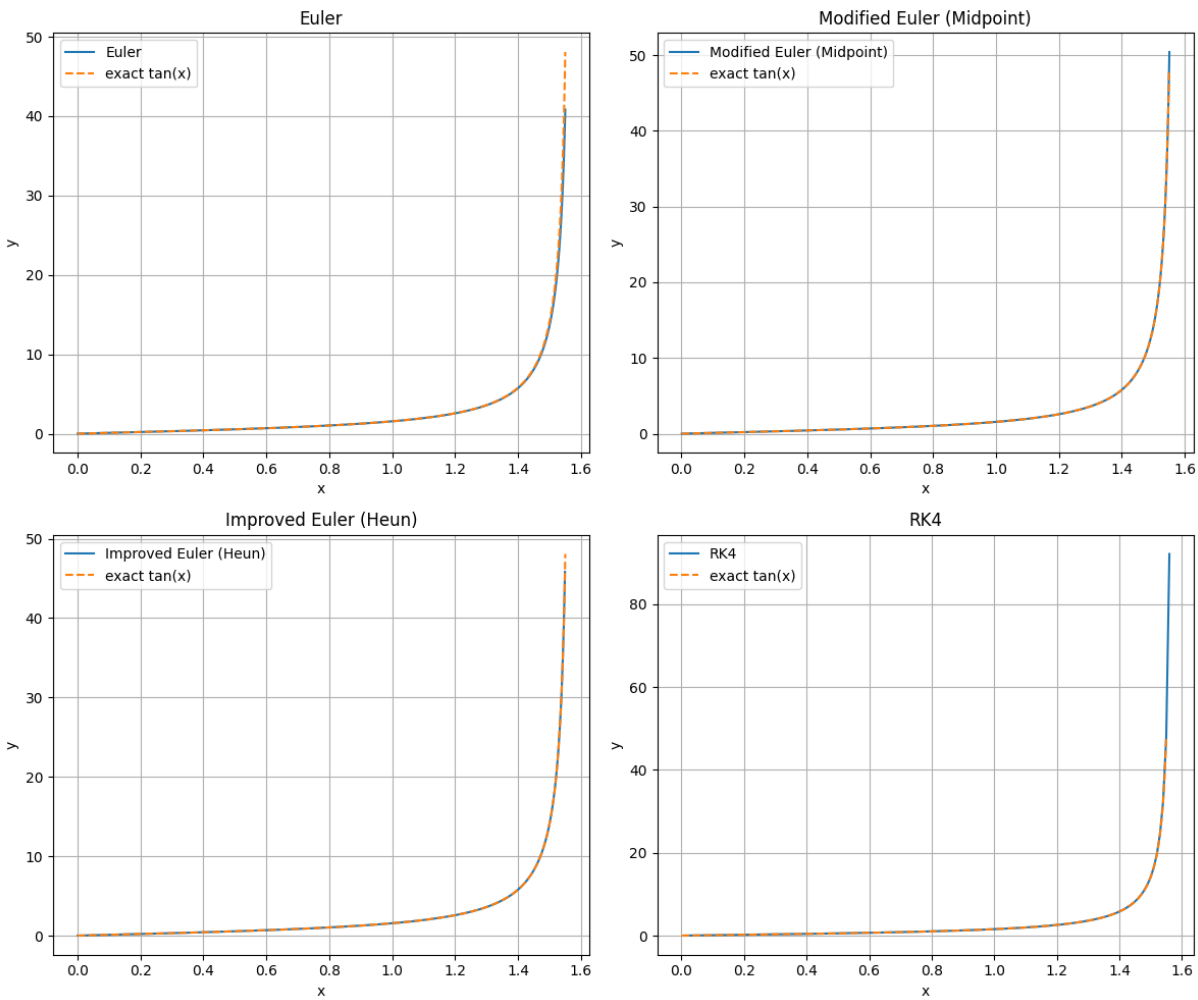


Figure 2: Comparison of numerical methods with true value

Results

Method	$y(1.55)$	**Error**
Euler (euler.dat)	40.8304780142	7.2481747173
Modified Euler (eulermod.dat)	48.0009453860	0.0775370932
Improved Euler (euler_impv.dat)	45.8272534837	0.0449258283
RK4 (rk4.dat)	48.0500659897	0.0283363712

Conclusion

The RK4 method provides the most accurate result with the smallest error even for larger updates and lesser iterations, while the simple Euler method shows the largest deviation from the true value.

Question 5-7

Nonlinear Oscillator Dynamics

Consider the nonlinear equation of motion:

$$\frac{d^2x}{dt^2} = -\sin(x)$$

with time step $dt = 0.01$, integrated for 5000 iterations (total time $T = 50$).

- Case 1: $x_0 = 0.1,\ v_0 = 1.9$
- Case 2: $x_0 = 0,\ v_0 = 1.999$
- Case 3: $x_0 = 0,\ v_0 = 2.4$

Compute the value of x at $t = 50$ and explain why the behavior changes significantly when $v_0 > 2$.

Results

Initial Conditions	$x(T = 50)$	**Behavior**
$x_0 = 0.1,\ v_0 = 1.9$	-2.3993325934678293	Oscillatory
$x_0 = 0,\ v_0 = 1.999$	-2.3412238000431116	Near-critical oscillation
$x_0 = 0,\ v_0 = 2.4$	91.17432446478745	Unbounded rotation

Energy Analysis

This system is equivalent to a simple pendulum. The total energy is:

$$E = \frac{1}{2}v^2 + (1 - \cos x)$$

Initial Energy

- Case 1:
$$E \approx \frac{1}{2}(1.9)^2 + (1 - \cos 0.1) \approx 1.805 + 0.005 = 1.81$$
- Case 2:
$$E = \frac{1}{2}(1.999)^2 \approx 1.998$$
- Case 3:
$$E = \frac{1}{2}(2.4)^2 = 2.88$$

Physical Interpretation

The potential energy is:

$$U(x) = 1 - \cos x$$

whose maximum value is:

$$U_{\max} = 2 \quad (\text{at } x = \pi)$$

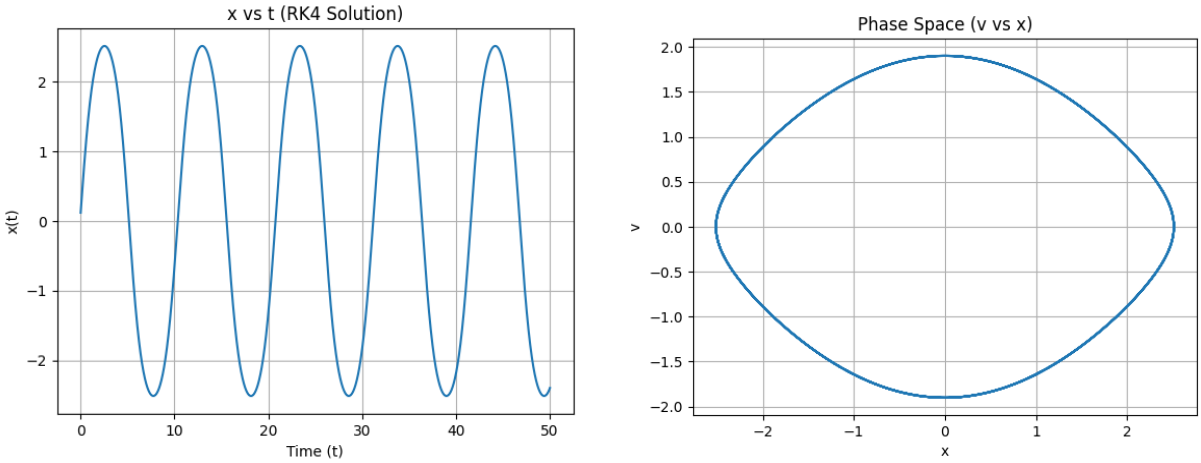
- If $E < 2$: The particle is trapped in the potential well $\Rightarrow$ **oscillatory motion**.
- If $E \approx 2$: The system is at the **separatrix**, showing slow, large oscillations.
- If $E > 2$: The particle escapes the well $\Rightarrow$ **rotational motion** (unbounded x).

Conclusion

- Case 1 ($E < 2$): Stable oscillations around equilibrium.
- Case 2 ($E \approx 2$): Critical behavior with large excursions.
- Case 3 ($E > 2$): Continuous rotation, leading to very large x values.

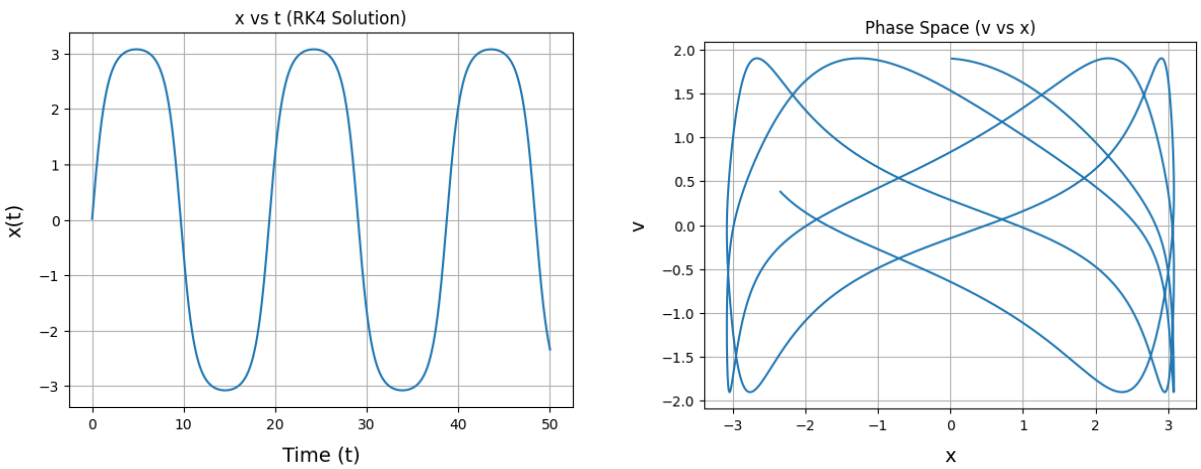
Thus, the qualitative change occurs because $E = 2$ is the critical energy separating bounded and unbounded motion.

Figures



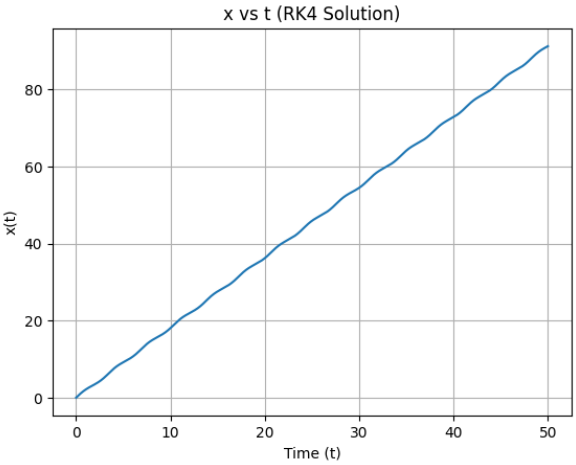
(a) x vs t (b) v vs x

Figure 3: Dynamics of stable oscillations

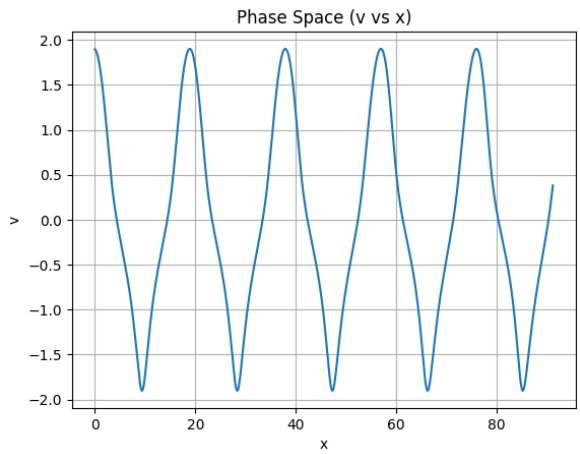


(a) x vs t (b) v vs x

Figure 4: Dynamics of critical oscillations



(a) x vs t



(b) v vs x

Figure 5: Dynamics of unbounded rotations

Question 8

Coupled Oscillator Ring (RK4)

Consider a system of 50 particles arranged on a circular ring of radius 5.0 in the x-z plane with periodic boundary conditions. Each particle can move only along the y-direction.

The force on the i-th particle is given by:

$$f_i^{(y)} = k_m \left(y_{i+1} + y_{i-1} - 2y_i\right)$$

where $k_m = \frac{\kappa}{m} = 1$.

Initial conditions:

- $v_i(0) = 0 \quad \forall i$
- $y_i(0) = 0 \quad \forall i$, except

$$y_1(0) = y_{26}(0) = 0.8$$

Task:

- Use the RK4 method with time step $dt = 0.02$
- Run for 2000 iterations ($t = 40$)
- Compute $y_i(t)$ for all particles

Find the displacement of selected particles at $t = 40$.

Results

Particle	$y_i(t = 40)$
y_1	-0.11891893262315607
y_{40}	-0.005328767596945476

Remarks

- The system behaves as a 1D lattice of coupled oscillators with periodic boundary conditions.
- The initial localized displacement splits into traveling waves propagating in both directions.
- Due to periodicity, waves interfere and redistribute energy across all particles.
- Over time, the motion becomes a superposition of normal modes.

Figure

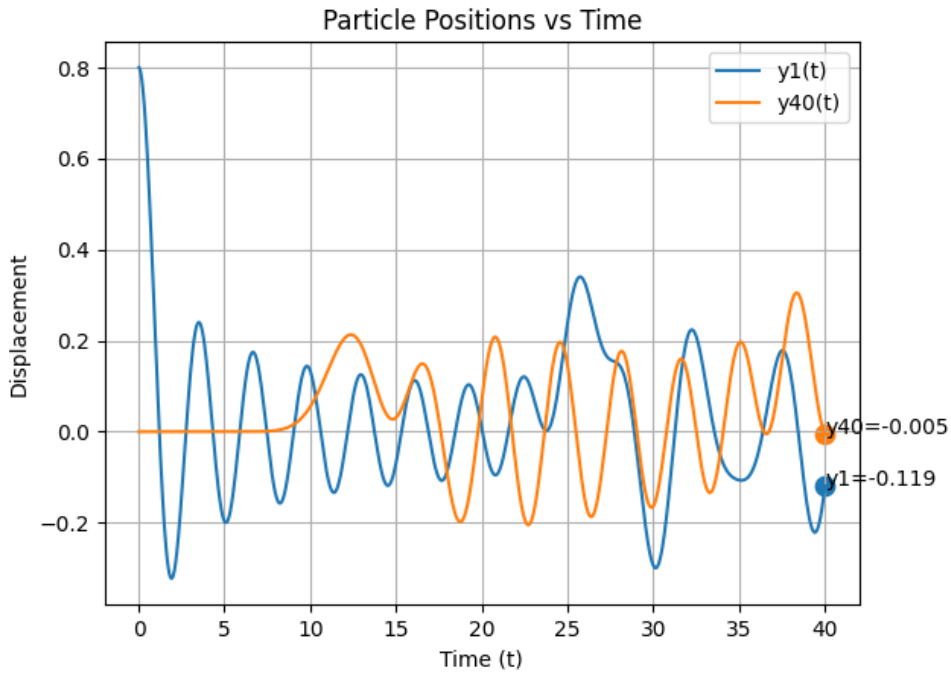


Figure 6: Coupled oscillator ring

Question 9

Boundary Value Problem using Gauss-Seidel Method

Solve the differential equation:

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 10y = 10x$$

on the interval $x \in [0, 1]$ using the Gauss-Seidel method.

Boundary conditions:

$$y(0) = 0, \quad y(1) = 2$$

Numerical details:

- Grid spacing: $dx = 0.01$
- Use double precision (real*8)
- Convergence criterion: 10^{-4}

Task: Compute the numerical solution and determine the value of y at $x = 0.80$.

Results

Quantity	Value
Number of grid points	100
Iterations for convergence	3912
$y(x = 0.80)$	0.36787301598290084

Remarks

- The differential equation is discretized using finite differences:

$$\frac{y_{i+1} - 2y_i + y_{i-1}}{dx^2} - 5\frac{y_{i+1} - y_{i-1}}{2dx} + 10y_i = 10x_i$$

- This leads to a linear system which is solved iteratively using the Gauss-Seidel method.
- Convergence is achieved when the maximum change in y_i across iterations is less than 10^{-4}.

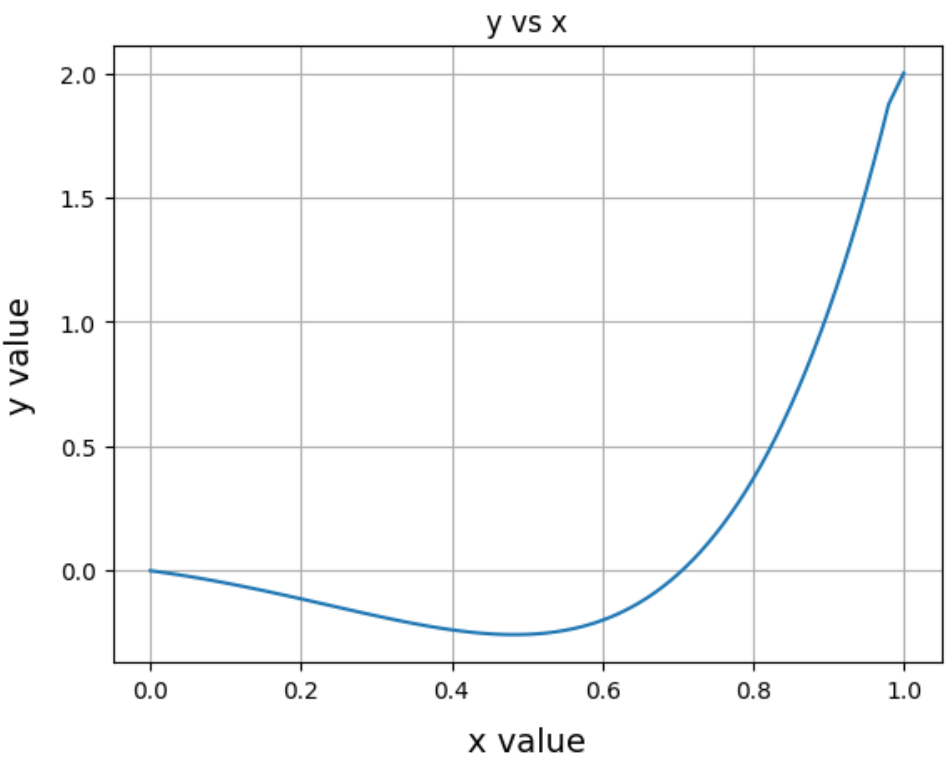


Figure 7: Gauss-Sidel method

Question 10

2D Laplace Equation on a Square Lattice

Consider a 34×34 2D plate. Solve Laplace's equation:

$$\nabla^2 T = 0$$

to obtain the steady-state temperature distribution on the lattice.

Boundary conditions:

- At $x = 1$: $T = 3.7$ for all y
- At $x = 34$: $T = 0.4$ for all y
- At $y = 1$: T decreases linearly from 3.7 to 0.4 as x goes from 1 to 34
- At $y = 34$: T decreases linearly from 3.7 to 0.4 as x goes from 1 to 34

Numerical details:

- Use an iterative method (e.g., Gauss-Seidel)

- Convergence criterion: 10^{-4}

Tasks:

- Compute the temperature at all lattice points
- Plot the temperature profile (heatmap) with properly labelled axes
- Find the temperature at the point $(20, 20)$

Result

$$T(20, 20) = 1.7791982400306057$$

Numerical Scheme

Laplace's equation in discrete form:

$$T_{i,j} = \frac{1}{4}\left(T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}\right)$$

Remarks

- The solution represents a steady-state heat distribution.
- Temperature smoothly interpolates between boundary values.
- The linear boundary conditions on top and bottom edges create a gradient across the plate.

Figure-Heatmap

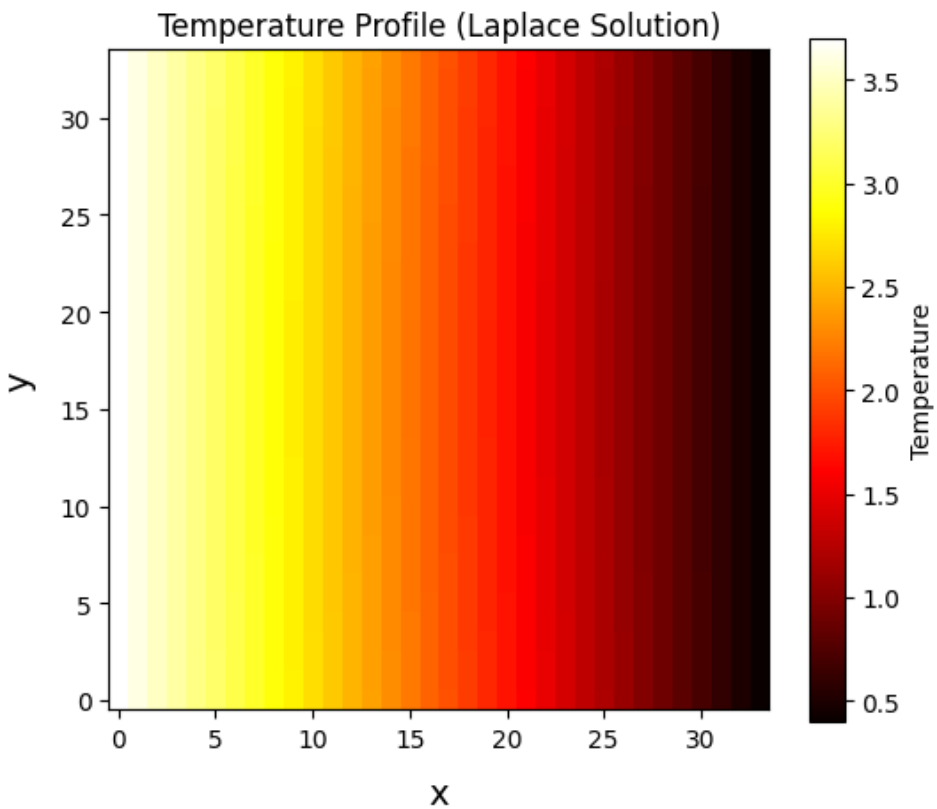


Figure 8: Temperature distribution on the 34×34 lattice

Question 11

2D Laplace Equation with Neumann Boundary Conditions

Consider a 34×34 2D plate. Solve Laplace's equation:

$$\nabla^2 T = 0$$

to obtain the steady-state temperature distribution on the lattice.

Boundary conditions (Neumann type):

- At $(1, 1)$: $T = 2000$
- Left boundary: $\dfrac{dT}{dx} = A = -70$
- Right boundary: $\dfrac{dT}{dx} = B = -40$
- Bottom boundary: $\dfrac{dT}{dy} = C = 20$
- Top boundary: $\dfrac{dT}{dy} = D = -10$

Numerical details:

- Use an iterative scheme (e.g., Gauss-Seidel)
- Convergence criterion: 10^{-5}

Task:

- Compute the temperature distribution on the lattice
- Determine the temperature at the point $(10, 10)$

Result

$$T(10, 10) = 1550.1274032835956$$

Figure (Heatmap Placeholder)

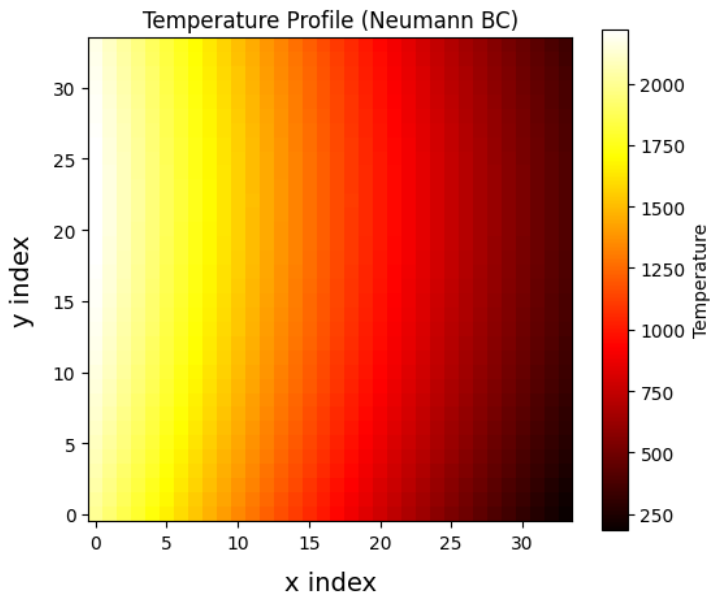


Figure 9: Temperature distribution with Neumann boundary conditions

Numerical Scheme

- Interior points satisfy:

$$T_{i,j} = \frac{1}{4}\left(T_{i+1,j} + T_{i-1,j} + T_{i,j+1} + T_{i,j-1}\right)$$

- Neumann boundary conditions are implemented using finite differences, e.g.:

$$\frac{T_{2,j} - T_{1,j}}{dx} = -70 \quad \Rightarrow \quad T_{1,j} = T_{2,j} + 70\,dx$$

Remarks

- Neumann conditions fix heat flux instead of temperature.
- The absolute temperature scale is anchored using $T(1,1) = 2000$.
- The solution reflects directional gradients imposed at the boundaries.